

Unit 1.2 - Sets and Functions

1 Sets

1.1 Definitions and Notation

- Sets are collections of like objects called *elements* or *members*
- Usually designated with capital letters: A, B, C
- Example: $A = \{1, 2, 3\}$
- Set membership: $1 \in A$ (1 is an element of A)
- Sets with one element (e.g., $\{1\}$) are called singletons
- The set with no elements ($\emptyset = \{\}$) is called the empty set
- Defining a set using a rule that its elements satisfy: $A = \{x \mid x > 0\}$ denotes the set of all x such that x is greater than 0.

1.2 Subsets

Let A and B be sets. We say that A is a **subset** of B (or A is contained in B) if every element of A is also an element of B . We denote this by writing $A \subseteq B$. In other words, $A \subseteq B$ if, for every element $x \in A$, $x \in B$ as well.

A set A is considered to be a **proper subset** of B if $A \subseteq B$ but $A \neq B$, meaning that A is contained in B but there is at least one element in B that is not in A . We denote this by writing $A \subset B$.

By convention, the **empty set** $\emptyset$ is considered a subset of all sets, since there are no elements in the empty set that could violate the condition of being in any set B . Thus, for any set B , we have $\emptyset \subseteq B$.

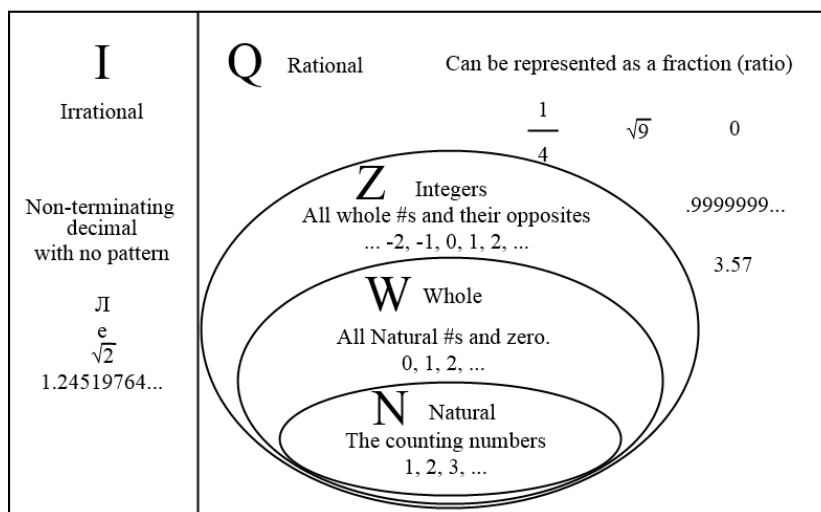
1.3 Sets of Numbers

There are several commonly used sets of numbers, including:

- Natural numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$ (sometimes including 0).
- Whole numbers: $\mathbb{W} = \{0, 1, 2, 3, \dots\}$ (usually including 0).
- Integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

- Rational numbers: $\mathbb{Q} = \{\frac{p}{q} \mid p, q \in \mathbb{Z} \text{ and } q \neq 0\}$.
 - Real numbers: $\mathbb{R}$ (including all rational and irrational numbers).
- Each of these sets is a subset of the next: $\mathbb{N} \subseteq \mathbb{W} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.

Real Numbers



1.4 Intervals

An **interval** is a set of real numbers that consists of all numbers lying between two given numbers, called the *end points* of the interval. Intervals are commonly used to describe contiguous subsets of the real number line.

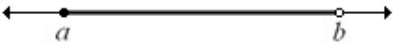
Intervals can be classified as either **open**, **closed**, or a combination of both, depending on whether or not the endpoints are included in the set.

1.4.1 Open Intervals

An **open interval** does not include its endpoints. The open interval from a to b (where $a < b$) consists of all real numbers x such that $a < x < b$. This is denoted by (a, b) , and is defined as:

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

In an open interval, neither endpoint a nor b is included in the set.

Interval Notation	Number Line Sketch	Set-builder Notation
(a, b)		$\{x \mid a < x < b\}$
$(a, b]$		$\{x \mid a < x \leq b\}$
$[a, b)$		$\{x \mid a \leq x < b\}$
$[a, b]$		$\{x \mid a \leq x \leq b\}$
(a, ∞)		$\{x \mid x > a\}$
$(-\infty, b)$		$\{x \mid x < b\}$
$[a, \infty)$		$\{x \mid x \geq a\}$
$(-\infty, b]$		$\{x \mid x \leq b\}$
$(-\infty, \infty)$		$\mathbb{R}$

1.4.2 Closed Intervals

A **closed interval** includes both of its endpoints. The closed interval from a to b (where $a < b$) consists of all real numbers x such that $a \leq x \leq b$. This is denoted by $[a, b]$, and is defined as:

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

In a closed interval, both endpoints a and b are included in the set.

1.4.3 Half-Open (or Half-Closed) Intervals

A **half-open** (or **half-closed**) interval includes one of its endpoints but not the other. There are two types of half-open intervals:

- $(a, b]$: The set of all x such that $a < x \leq b$. This interval includes b but not a .
- $[a, b)$: The set of all x such that $a \leq x < b$. This interval includes a but not b .

Infinite Intervals

Intervals may also extend infinitely in one or both directions. Examples include:

- (a, ∞) : The set of all $x \in \mathbb{R}$ such that $x > a$. This is an open interval extending to infinity.
- $(-\infty, b]$: The set of all $x \in \mathbb{R}$ such that $x \leq b$. This is a half-closed interval extending to negative infinity.
- $(-\infty, \infty)$: The set of all real numbers $\mathbb{R}$. This is an open interval extending infinitely in both directions.

1.5 Set Operations

1.5.1 Union

The **union** of two sets A and B , denoted by $A \cup B$, is the set that contains all elements that are in A , in B , or in both. The union operation combines the elements of both sets without duplication. Formally, the union is defined as:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

1.5.2 Intersection

The **intersection** of two sets A and B , denoted by $A \cap B$, is the set that contains all elements that are both in A and B . The intersection operation finds the common elements between the two sets. Formally, the intersection is defined as:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

If two sets have no elements in common (i.e., $A \cap B = \emptyset$), then the sets are **disjoint**.

1.5.3 Complement (or Difference)

The **complement** (or **set difference**) of set B relative to set A , denoted by $A \setminus B$, is the set of all elements that are in A but not in B . Formally, the set difference is defined as:

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

This operation removes all elements of set B from set A .

1.5.4 Examples

If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$, then:

- $A \cup B = \{1, 2, 3, 4\}$
- $A \cap B = \{2, 3\}$
- $A \setminus B = \{1\}$

- $B \setminus A = \{4\}$

Now for some examples with intervals:

- **Union of Intervals:** $(1, 3) \cup [2, 5] = (1, 5]$

Explanation: The union of the open interval $(1, 3)$ and the closed interval $[2, 5]$ includes all numbers from 1 to 5, with 1 not included (since it is an open interval on the left) and 5 included (since it is a closed interval on the right).

- **Intersection of Intervals:** $(1, 3) \cap [2, 5] = [2, 3]$

Explanation: The intersection of the open interval $(1, 3)$ and the closed interval $[2, 5]$ includes all numbers that are in both intervals, which are those from 2 to 3, INCLUDING 2 and 3 (since both of these endpoints are within both of the original intervals).

- **Complement (Set Difference) of Intervals:** $[1, 4] \setminus (2, 3) = [1, 2] \cup [3, 4]$

Explanation: The complement of the open interval $(2, 3)$ within the closed interval $[1, 4]$ is the set of all numbers in $[1, 4]$ that are not in $(2, 3)$. This results in two separate intervals: from 1 to 2 (including 1 and 2), and from 3 to 4 (including both endpoints).

- **Union of Infinite Intervals:** $(-\infty, 2] \cup (3, \infty) = (-\infty, 2] \cup (3, \infty)$

Explanation: The union of two infinite intervals $(-\infty, 2]$ and $(3, \infty)$ includes all numbers less than or equal to 2 and all numbers greater than 3. There is no overlap between these intervals, so the union is simply the combination of both intervals.

- **Intersection of Infinite Intervals:** $(-\infty, 4) \cap (2, \infty) = [2, 4]$

Explanation: The intersection of the two infinite intervals $(-\infty, 4)$ and $(2, \infty)$ is the set of all numbers that are both less than or equal to 4 and greater than or equal to 2. This results in the closed interval $[2, 4]$.

- **Complement of an Interval:** $\mathbb{R} \setminus [1, 3] = (-\infty, 1) \cup (3, \infty)$

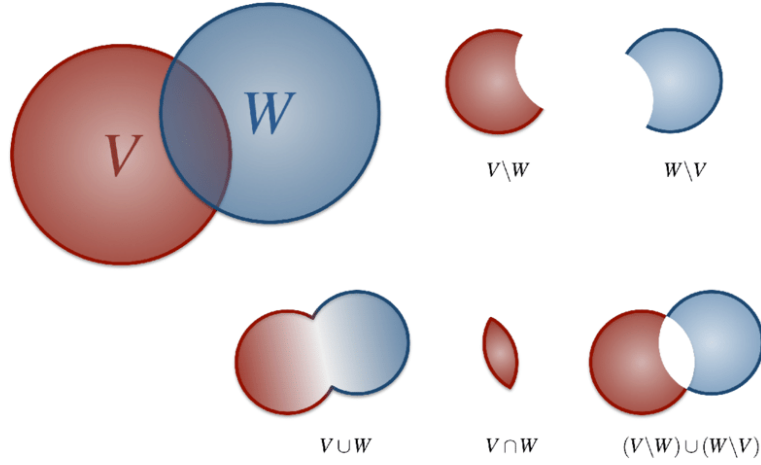
Explanation: The complement of the closed interval $[1, 3]$ in the set of all real numbers $\mathbb{R}$ consists of all numbers less than 1 or greater than 3. Thus, the result is the union of two infinite intervals: $(-\infty, 1)$ and $(3, \infty)$.

1.5.5 Properties

The union and intersection operations satisfy several important properties:

- **Commutative Property:**

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$



- **Associative Property:**

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

- **Distributive Property:**

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

1.6 Union and Intersection of Indexed Collections of Sets

An **indexed collection of sets** $A_1, A_2, \dots, A_n$ is a family of sets that are labeled by an index, usually $1, 2, 3, \dots, n$.

1.6.1 Union of Indexed Collections of Sets

The **union** of an indexed collection of sets is the set that contains all elements that belong to at least one of the sets. It is denoted by:

$$\bigcup_{i \in I} A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

1.6.2 Intersection of Indexed Collections of Sets

The **intersection** of an indexed collection of sets is the set that contains all elements that are common to every set. It is denoted by:

$$\bigcap_{i \in I} A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

1.6.3 Examples

Let $I = \{1, 2, 3\}$ and consider the indexed collection of sets $\{A_i\}_{i \in I}$ where:

- $A_1 = \{1, 2\}$
- $A_2 = \{2, 3\}$
- $A_3 = \{2, 4\}$

Then:

- $\bigcup_{i=1}^3 A_i = A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 4\}$
- $\bigcap_{i=1}^3 A_i = A_1 \cap A_2 \cap A_3 = \{2\}$

1.7 Tuples and Ordered Pairs

1.7.1 Ordered Pairs

An **ordered pair** is a collection of two elements in which the order of the elements matters. An ordered pair is denoted by (a, b) , where a is the first element and b is the second element (NOTE: be aware that this is the same notation as open intervals and that context should let you know what is meant). Two ordered pairs (a, b) and (c, d) are considered equal if and only if $a = c$ and $b = d$.

Ordered pairs are fundamental in defining Cartesian coordinate systems, relations, and functions. For example, the point $(2, 3)$ represents an ordered pair in a Cartesian coordinate system, where 2 is the x -coordinate, and 3 is the y -coordinate.

1.7.2 Tuples

A **tuple** is a generalization of an ordered pair that can contain any finite number of elements, and the order of these elements is important. An n -tuple is a sequence of n elements, denoted by $(a_1, a_2, \dots, a_n)$, where each a_i represents an element in the tuple.

- A **2-tuple** is also known as an **ordered pair**, such as (a, b) .
- A **3-tuple** (or **triple**) contains three elements, e.g., (a, b, c) .
- A **4-tuple** contains four elements, e.g., (a, b, c, d) .
- In general, an n -**tuple** contains n elements.

1.7.3 Equality of Tuples

Two n -tuples $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$ are equal if and only if their corresponding elements are equal, i.e., $a_i = b_i$ for all $i = 1, 2, \dots, n$.

1.7.4 Examples

- The ordered pair $(2, 5)$ is different from the ordered pair $(5, 2)$ because the order of the elements matters.
- The 3-tuple $(1, 2, 3)$ is different from the 3-tuple $(3, 2, 1)$ because the order of the elements is different.
- In a Cartesian coordinate system, the point $(x, y, z) = (4, -2, 7)$ represents a location in three-dimensional space.

2 Functions

2.1 Relations

A **relation** R between two sets A and B is a rule that associates elements of A with elements of B . More formally, a relation is a set of ordered pairs (a, b) where $a \in A, b \in B$. Intuitively, a relation R from set A to set B can be thought of as a collection of connections or links between elements of A and elements of B . If $(a, b) \in R$, we say that a is related to b by the relation R .

- Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$. A relation R from A to B could be:

$$R = \{(1, x), (2, x), (3, y)\}.$$

This relation R consists of the set of ordered pairs where:

- 1 is related to x ,
 - 2 is related to x , and
 - 3 is related to y .
- Let $A = \mathbb{R}$ (the set of all real numbers) and $B = \mathbb{R}$. The relation R defined by “ x is less than y ” can be expressed as:

$$R = \{(x, y) \mid x, y \in \mathbb{R}, x < y\}.$$

This relation R consists of all ordered pairs (x, y) such that x is a real number less than y .

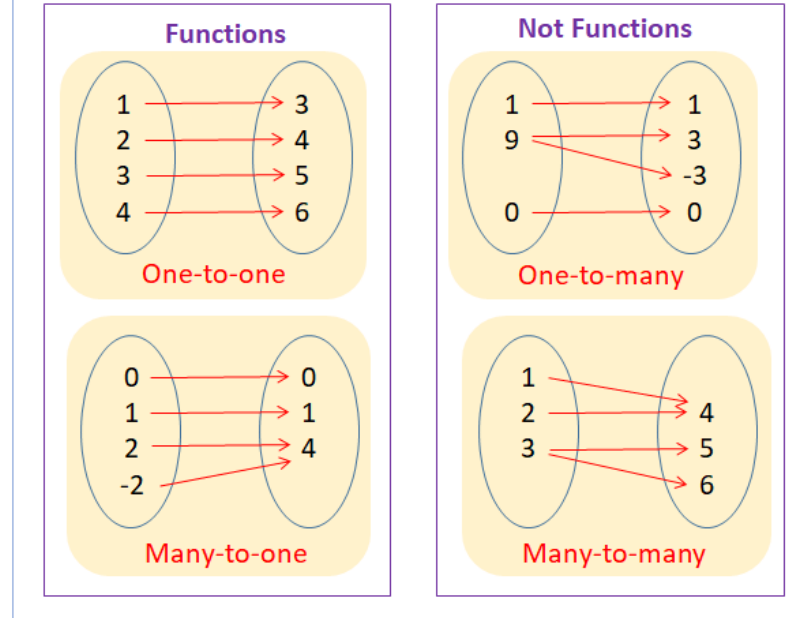
2.2 Functions

A **function** is a special type of relation from a set A (called the *domain*) to a set B (called the *range*) such that every element $x \in A$ is related to exactly one element $y \in B$. In other words, a function is a relation where each input from the domain A is related to a single output in the codomain B . However, the same output in B can be related to multiple inputs from A . This means a function is *many-to-one* but not *one-to-many*. Functions are *one-to-one* if both the inputs AND outputs are uniquely related to single elements.

Relations

A relation shows a relationship between two values.

A function is a relation where each input has only one output.



2.2.1 Notation

Functions are typically denoted by letters such as f , g , or h . The notation $f : A \rightarrow B$ means that f is a function from set A to set B . If $f(x) = y$, then x is called the *input* or *argument* of the function, and y is the *output* or *value* of the function at x .

2.2.2 Examples of Functions

- Consider the function $f : \{1, 2, 3\} \rightarrow \{a, b\}$ defined by:

$$f(1) = a, \quad f(2) = a, \quad f(3) = b.$$

Here, every element in the domain $\{1, 2, 3\}$ is assigned exactly one element in the codomain $\{a, b\}$. The range of f is $\{a, b\}$.

- Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $g(x) = x^2$. In this case:
 - The domain of g is $\mathbb{R}$, the set of all real numbers.
 - The range of g is $[0, \infty)$, the set of all non-negative real numbers, since $x^2 \geq 0$ for all $x \in \mathbb{R}$.

- Let $h : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $h(n) = n + 1$. Here:
 - The domain of h is $\mathbb{Z}$, the set of all integers.
 - The range of h is also $\mathbb{Z}$, since for every integer n , $n + 1$ is also an integer.

2.2.3 Finding the Domain and Range of Real-Valued Functions

A *real-valued function* is a function whose range is the set of real numbers.

- **Example 1:** $f(x) = \sqrt{x - 2}$
 - **Domain:** The function $f(x)$ involves a square root, which is only defined for non-negative numbers. Therefore, we need $x - 2 \geq 0$, which gives $x \geq 2$. Thus, the domain of f is $[2, \infty)$.
 - **Range:** Since the square root function outputs only non-negative values, the range of $f(x) = \sqrt{x - 2}$ is $[0, \infty)$.
- **Example 2:** $g(x) = \frac{1}{x-3}$
 - **Domain:** The function $g(x)$ involves a division by $x - 3$. Division by zero is undefined, so we require $x - 3 \neq 0$, which gives $x \neq 3$. Thus, the domain of g is $\mathbb{R} \setminus \{3\}$.
 - **Range:** The function $g(x) = \frac{1}{x-3}$ can take any real value except 0, as there is no x for which $\frac{1}{x-3} = 0$. Thus, the range of g is $\mathbb{R} \setminus \{0\}$.
- **Example 3:** $h(x) = \ln(x + 1)$
 - **Domain:** The natural logarithm function $\ln(x)$ is only defined for positive inputs. Thus, we need $x + 1 > 0$, which gives $x > -1$. Therefore, the domain of h is $(-1, \infty)$.
 - **Range:** The range of the natural logarithm function is all real numbers, so the range of $h(x) = \ln(x + 1)$ is $\mathbb{R}$.

2.3 Functions in Machine Learning

In machine learning, functions play a fundamental role in modeling the relationships between data inputs (often called *predictors* or *features*) and outputs (often called *labels* or *targets*). The goal of many machine learning algorithms is to find a function $f : X \rightarrow Y$ that maps each input $x \in X$ to a corresponding output $y \in Y$.

Machine learning methods are designed to automatically learn such functions from data. Given a set of training data, which consists of pairs (x_i, y_i) where x_i represents the features and y_i represents the labels (and i is an index which runs over the number of data samples you have), the learning algorithm attempts to find a function $f(x)$ that best describes the relationship between the inputs and outputs.